

Understanding the Decision-Making Process in Brain Tumor Classification: A Comparative Grad-CAM Analysis of ResNet50, VGG16, EfficientNetB3, DenseNet121, and InceptionV3

1st Calvin Suharjono
School of Computer Science
Bina Nusantara University
Tangerang, Indonesia
calvin.suharjono@binus.ac.id

2nd Ringo Gary Buntono
School of Computer Science
Bina Nusantara University
Tangerang, Indonesia
ringo.buntono@binus.ac.id

3rd Stefanus Abel Filio
School of Computer Science
Bina Nusantara University
Tangerang, Indonesia
stefanus.filio@binus.ac.id

Abstract

Early detection of brain tumors is a very important measure to increase the probability of patient survival. However, interpreting the results manually can be time consuming and prone to errors, therefore using artificial intelligence is expected to be able to detect quickly and efficiently. On this occasion, researchers used 5 pretrain models of CNN architecture, namely models—ResNet50, VGG16, EfficientNetB3, DenseNet121, and InceptionV3—using the Grad-CAM technique to visualize model decision-making processes. A dataset of 7023 MRI images was used for classification (glioma, meningioma, pituitary, and healthy), with preprocessing steps including resizing, grayscale conversion, contour-based cropping, and data augmentation. Each model was fine-tuned and evaluated using accuracy, precision, recall, F1-score, and confusion matrices, while Grad-CAM heatmaps assessed model interpretability. The results show that all five models produce, with DenseNet121 and EfficientNetB3 performing best in terms of accuracy and generalization. The Resnet 50 model shows a stable model and the Inception v3 model shows the best model for interpretability because it has the largest average drop. This study shows that good accuracy does not necessarily result in good interpretability.

Keywords : CNN , MRI, ResNet , VGG ,EfficientNet, DenseNet, Inception, Grad-Cam, Average Drop.

I. Introduction

Brain tumors are among the most dangerous health conditions, and early, accurate diagnosis is essential for improving patient survival. MRI is commonly used for detecting brain tumors because it offers high-quality images and is non-invasive. However, analyzing MRI scans manually is time-consuming and can lead to errors, which creates a strong need for automated deep learning-based classification systems.

Convolutional Neural Networks (CNNs) have proven highly effective in image classification, including in medical imaging. However, a major challenge is the

limited availability of large medical datasets, which makes training CNNs from scratch difficult. To solve this, transfer learning has become a valuable approach by using pre-trained models like ResNet, VGG, EfficientNet, DenseNet, and Inception. These models, trained on large image datasets, help improve performance and speed in medical tasks with limited data.

In medical applications, interpretability is critical because healthcare professionals must understand how and why an AI model makes its decisions. This study uses Grad-CAM (Gradient-weighted Class Activation Mapping) to produce a heat map showing which parts the model considers important for making a decision.

This study shows how effective the use of grad cam is in interpreting CNN models—ResNet, VGG, EfficientNet, DenseNet, and Inception—to find out which model provides the best interpretation in brain tumor classification, and help build trust in AI as a tool to assist medical professionals.

II. Literature Review

Brain tumor classification is an important task to do. Usually the data is taken from mri scan and ct scan with MRI being the preferred choice due to its superior contrast resolution for soft tissues. However, accurate and rapid diagnosis remains a challenge due to the many different types of tumors and the need for experienced specialists. Deep learning, especially CNN, has shown success in detecting various diseases around the world which use medical images. CNN excels in extracting spatial hierarchies of features, making CNN effective in tumor classification tasks by learning complicated patterns and reducing the reliance on manual feature engineering. Studies like Pereira et al. (2016) and Afshar et al. (2020) shows the effectiveness of the cnn model in classifying brain tumors using a dataset of medical mri images.

The last few years, famous Convolutional Neural Network (CNN) architectures like ResNet50, VGG16,

EfficientNetB3, DenseNet121, and InceptionV3 have been widely used for disease classification, one of which is brain tumors. **VGG16**, while structurally simple and widely adopted, suffers from a large number of parameters and slower training times, making it less efficient for real-time applications [Haq et al., 2021]. **ResNet50** effectively handles deeper network training through residual connections, which help mitigate vanishing gradient problems and enhance classification performance on complex medical images [Chowdhury et al., 2021]. **EfficientNetB3** has emerged as a highly accurate and computationally efficient model by applying compound scaling to network depth, width, and resolution [Zhou et al., 2022]. **DenseNet121** offers strong feature propagation and reuse by densely connecting layers, which is particularly useful for medical datasets with limited samples [Naser & Deen, 2020]. Meanwhile, **InceptionV3** continues to perform well in medical tasks due to its factorized convolutions that reduce computational cost while preserving accuracy, although its complex design can pose challenges in customization [Agarwal & Pandey, 2021]. Each model of course has its own advantages and disadvantages which can affect the accuracy and time complexity of the model train.

Explainable Artificial Intelligence (XAI) becomes important especially in high-risk applications such as medical diagnosis, where understanding model decisions can impact patient outcomes. As deep learning models like Convolutional Neural Networks (CNNs) operate as "black boxes," professionals such as doctors need a tool that can be used to increase their confidence in the prediction results and transparency of the MRI photo results. One of the techniques or methods for XAI technique is **Gradient-weighted Class Activation Mapping (Grad-CAM)**, which provides a visual that displays a heat map that shows which parts of the dataset the model considers important for models to make a decision. Grad-CAM works by utilizing the gradients of the target class flowing into the final convolutional layer to produce a coarse localization map, indicating areas of interest in the input [Selvaraju et al., 2017]. This technique has proven particularly useful in medical imaging tasks, such as brain tumor classification, where it allows experts to validate whether the model is focusing on relevant anatomical structures [Lamy et al., 2022]. The result is that Grad-Cam is a tool used for interpretability to explain unclear black box models.

Several studies have used CNN architectures to see how well these CNN models perform for brain tumor classification tasks. For example, Pereira et al. (2020) looked at how well VGG16, ResNet50, and InceptionV3 worked on brain MRI images and found that ResNet50 was the most accurate because it could keep spatial information through skip connections. In the same way, Haq et al. (2021) tested CNNs like DenseNet121 and VGG16 to see how well they could classify brain tumors. They found that DenseNet121 did better with fewer parameters and better feature reuse. A newer study by Mohan et al. (2023) looked at different models (like EfficientNetB3, ResNet50, and MobileNetV2) for classifying gliomas and found that EfficientNetB3 was the

best because it was the most accurate and took the least amount of time to make predictions. This means it can be used in clinical settings. These comparisons are very important for helping to choose the right model for real-world medical AI systems.

Researchers use explainable AI (XAI) techniques such as Grad-Cam to show or understand how models classify brain tumors. This helps people to understand how the model works and builds trust in the artificial intelligence model. For example, Afshar et al. (2020) used Grad-CAM to show the important areas that affect a Capsule Network's predictions on brain MRI scans. This showed that the model was focusing on areas that were affected by tumors. In the same way, Kumar et al. (2021) added Grad-CAM to a CNN-based classification pipeline and found that visualizations helped radiologists check their predictions and find mistakes in their classifications. These studies show that interpretability is becoming more and more important in medical AI, especially in fields where people need to be very accountable. Even though there has been a lot of progress, most of the best deep learning methods right now are mostly focused on making models more accurate, with little thought given to how easy they are to use in clinical settings. Also, a lot of models are trained and tested on small or unbalanced datasets, which makes it hard to know if they will work in other situations. XAI techniques like Grad-CAM can be used as both validation tools and feedback mechanisms for improving models over time. This is especially important for multi-class or multi-modal brain tumor classification tasks where being able to understand the results is still very important for making decisions.

III. Methodology

a. Dataset

In this study, the "Brain Tumor MRI Scans" dataset obtained from the Kaggle platform and provided by Rajarshi Mandal was used. This dataset consists of 7023 brain MRI images classified into four categories, namely glioma, meningioma, pituitary, and healthy. This data is divided into two parts, namely training data of 5712 images and test data of 1311 images. This dataset was chosen because it has a fairly large amount of data and covers a variety of types of brain tumors that are commonly found, so it is considered suitable to support the training and testing process of the image classification model in this study.

b. Preprocessing

In the data preprocessing stage, researchers take several steps to prepare the dataset before it is used in the model training process. The first step is to resize each image in the dataset to 224 x 224 pixels. This adjustment aims to ensure that all images have uniform dimensions, thus facilitating the subsequent processing process. After the image size is adjusted, the image is then converted into grayscale format to simplify visual information that is considered less influential in the tumor detection process, such as color information.

The next step is to detect the largest contour in each grayscale image. The largest contour is assumed to be the main area that represents the presence of a tumor. Based on these contours, the system identifies the extreme points, namely the top, bottom, left, and right points of the object in the image. These points are used as a reference for the image cropping or cutting process, so that only the relevant parts are retained. The cropped image is then saved in a new folder and used as input in the classification model training stage.

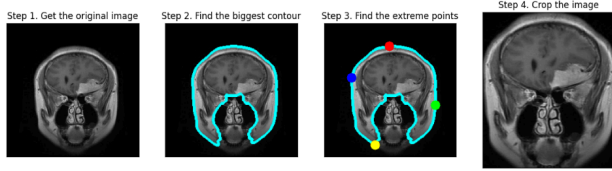


Figure 1. Image processing steps: (1) Original image, (2) Finding the largest contour, (3) Identifying the extreme points, and (4) Cropping the image based on the contour.

c. Data Augmentation

To increase data variation and prevent overfitting in the model, researchers applied data augmentation techniques using the ImageDataGenerator function from the Keras library. This augmentation includes several simple transformations to the image, such as random rotation up to 10 degrees (`rotation_range=10`), horizontal and vertical position shifts of 5% of the image dimension each (`width_shift_range=0.05` and `height_shift_range=0.05`), and horizontal image flips (`horizontal_flip=True`). By applying this technique, the model can learn from a wider variety of images, even though they come from the same original data, so it is expected to be able to improve model generalization when faced with test data that has never been seen before.

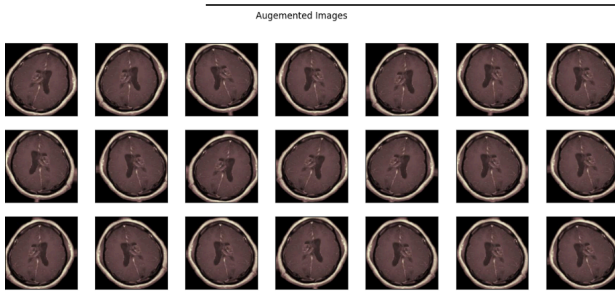


Figure 2. Sample of augmented brain tumor images using various image augmentation techniques

d. Model Architecture

In this study, researchers used several pre-trained models such as ResNet50, InceptionV3, DenseNet121, VGG16, and EfficientNetB3 as the base architecture for brain tumor image classification. All models were configured without including the top layer (`include_top=False`) to be tailored to specific classification tasks. These models used pre-trained weights on the ImageNet dataset to leverage the knowledge gained from large-scale data. In addition, all layers of the base model were enabled (`trainable=True`) so that they could be retrained to

improve performance on the dataset used. At the top of the model, several custom layers were added consisting of GlobalAveragePooling2D to smooth the feature output, Dropout to prevent overfitting, and Dense with softmax activation function as the final classification layer for multi-class classification tasks.

e. Grad-CAM Implementation

In this study, we used Gradient-weighted Class Activation Mapping (Grad-CAM) to increase the credibility and interpretability of the model, especially for multiclass brain tumor classification tasks. Using this technique, it allows us to see how the model makes decisions by displaying a heat map that shows which parts the model considers important.

1) Process of Grad-CAM Implementation

We apply the last convolutional layer of the model to extract the feature maps. These feature maps contain the learned representations of the input image at various levels of abstraction. To compute Grad-CAM, we first obtain the gradients of the predicted class score with respect to these feature maps.

The task of Grad-CAM is to compute the average gradient across all feature maps. This step is done to give weight to each feature map. This is important to do to predict the target class.

Once the gradients are computed, we weight the feature maps with these gradients and sum them together to generate a weighted combination of the feature maps. This weighted sum highlights the regions of the image that were most important for the model's decision.

A ReLU activation is applied to the weighted sum to ensure that only positive influences on the decision are considered, eliminating negative contributions. The resulting activation map is then upsampled to match the dimensions of the input image and overlaid as a heatmap.

2) Visualizing the Heatmap

The generated heatmap is overlaid on the original MRI image to visually show the areas that the model focused on when making its prediction. In the context of brain tumor classification, the heatmap typically highlights areas corresponding to the tumor regions in the MRI, providing evidence that the model is indeed focusing on the correct areas of the image.

3) Evaluation of Grad-CAM

The Grad-CAM visualizations are evaluated qualitatively by comparing them to the ground truth annotations (manual tumor segmentation by radiologists). The accuracy of these visualizations helps in confirming that the model is learning relevant features and focusing on clinically significant areas, such as the tumor location. Additionally, we compare the performance of Grad-CAM across different models (ResNet50, InceptionV3, and

EfficientNet) to determine which model provides the most reliable and interpretable results.

f. Model Evaluation

To evaluate our models, we use evaluation metrics used to assess model performance in brain tumor classification. We compare several models, namely ResNet50, VGG16, EfficientNetB3, DenseNet121, and InceptionV3, based on various performance metrics, such as accuracy, F1-score, precision, and confusion matrix.

1) Accuracy

Accuracy is the ratio of the number of correct predictions to the total number of predictions made. It is one of the simplest and most commonly used metrics to evaluate classification models. However, accuracy can be misleading when dealing with imbalanced datasets (where one class significantly outnumbers the others).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2) Recall

Recall is the ratio of correctly predicted positive instances to all actual positive instances in the dataset. This metric evaluates how well a model can identify all relevant cases within a specific class. **Recall is especially important when the goal is to minimize false negatives**, such as in medical diagnosis, fraud detection, or other high-risk scenarios where missing a positive case can have serious consequences.

$$Recall = \frac{TP}{TP + FN}$$

3) Precision

Precision measures how accurate the positive predictions are. It is the proportion of true positive results among all positive predictions made by the model. Precision is important when the cost of false positives is high (e.g., predicting a disease when it's not present).

$$Precision = \frac{TP}{TP + FP}$$

4) F1-score

The F1-score is the harmonic mean of precision and recall. It is especially useful when you need a balance between precision and recall, and when you have imbalanced data. A high F1-score indicates that both false positives and false negatives are low.

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall}$$

5) Confusion Matrix

A confusion matrix is a table that describes the performance of a classification model by showing the correct and incorrect predictions broken down by each class. For binary classification, it consists of four components: True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

6) Average Drop

Average Drop is a quantitative metric used to measure how much a model's confidence has decreased, that is produced by Grad-Cam. A low average drop value indicates that the model is stable but the model is unsuccessful in determining important locations in a model.

$$AverageDrop = \frac{1}{N} \sum_{i=1}^N \frac{\max(0, Si(x) - Si(x'))}{Si(x)} \times 100\%$$

IV. Results and Discussion

a. Results

1) ResNet50

This ResNet50 model is successful in brain tumor classification of four classes of brain tumors based on accuracy, evaluation metrics, and confusion metrics. The training and validation accuracy curves show a steady increase since the beginning of training, with validation accuracy reaching more than 95% after 20 epochs, indicating good generalization ability without any indication of overfitting. Based on the evaluation table, the model records high precision, recall, and f1-score across all classes, with the best performance in the no_tumor class achieving a perfect score. This model is also evaluated using a confusion matrix, which can be seen in figure 5, in the table it can be seen that the horizontal part is blue, this shows that the model has succeeded in classifying multi-classes well even though there are some errors. Overall, the model we have created has achieved good performance for brain tumor classification, this can be proven by several evaluation methods that we have shown below.

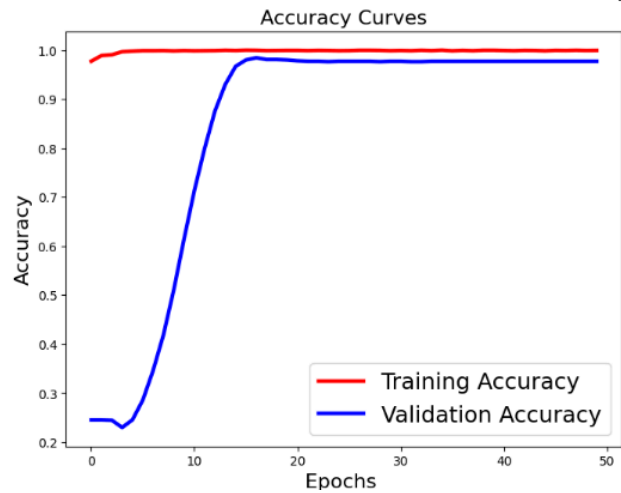


Figure 3. Training and validation accuracy curves of the ResNet50 model.

	Precision	Recall	f1-score
glioma	0.99	0.98	0.98
meningioma	0.98	0.97	0.98

no tumor	0.99	1.00	1.00
pituitary	0.98	0.99	0.99

Figure 4. Precision, Recall, and F1-score of the ResNet50 model.

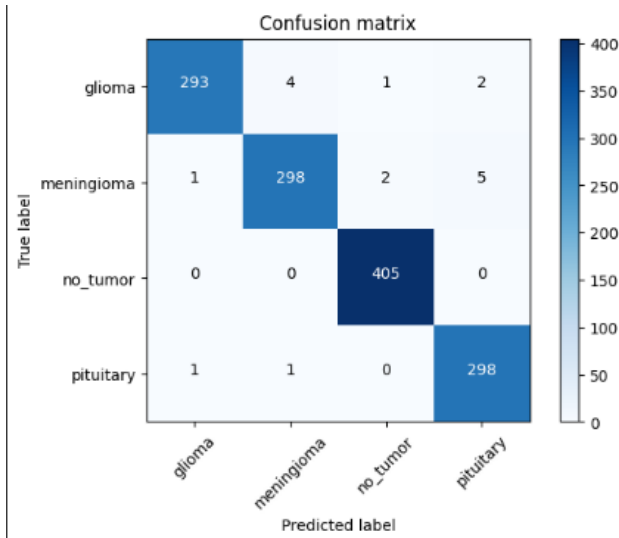


Figure 5. Confusion Matrix of the ResNet50 model.

2) InceptionV3

This InceptionV3 model is successful in brain tumor classification of four classes of brain tumors based on accuracy, evaluation metrics, and confusion metrics. The accuracy curve of the Inception v3 model shows a rapid increase, this can be seen from the 10th epoch, the model accuracy has stabilized at 97%. This shows that the model is able to learn the dataset perfectly without any significant overfitting. From the evaluation table, it can be seen that the model gets good precision, recall, and f1-score. With perfect values (1.00) in the no_tumor and pituitary classes, and values of 0.98 in the glioma and meningioma classes. This model is also evaluated using a confusion matrix, which can be seen in figure 8, in the table it can be seen that the horizontal part is blue, this shows that the model has succeeded in classifying multi-classes well even though there are some errors. Overall, the model we have created has achieved good performance for brain tumor classification, this can be proven by several evaluation methods that we have shown below.

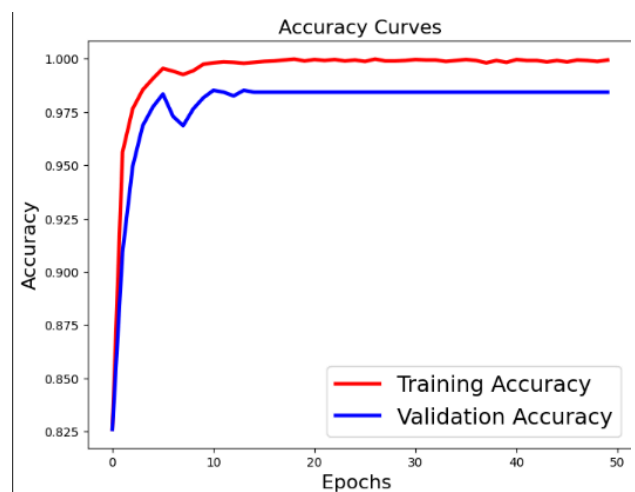


Figure 6. Training and validation accuracy curves of the InceptionV3 model.

	Precisi on	Recall	f1-score
glioma	0.99	0.97	0.98
meningioma	0.98	0.98	0.98
no tumor	1.00	1.00	1.00
pituitary	0.99	1.00	0.99

Figure 7. Precision, Recall, and F1-score of the InceptionV3 model.

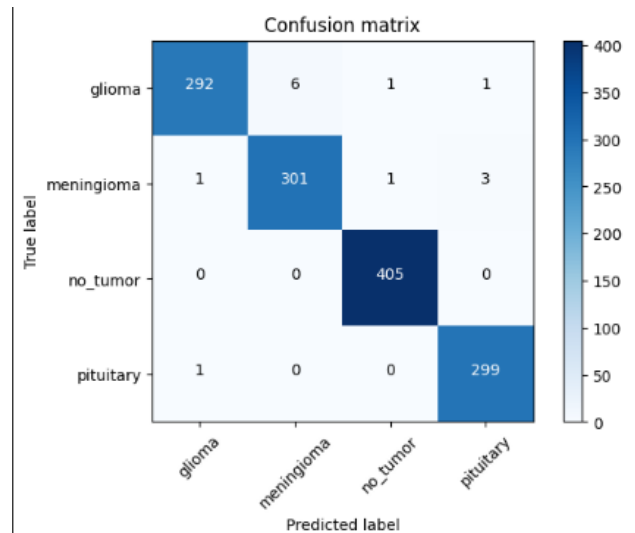


Figure 8. Confusion Matrix of the InceptionV3 model.

3) DenseNet121

This DenseNet121 model is successful in brain tumor classification of four classes of brain tumors based on accuracy, evaluation metrics, and confusion metrics. The accuracy curve of the DenseNet121 model shows a rapid increase, this can be seen from the 10th epoch, the model accuracy has stabilized at 99%. The evaluation results using precision, recall, and f1-score metrics for four

classes are also very satisfactory, with all metric values ranging from 0.97 to 1.00. The best performance is shown in the no_tumor class, which obtains a perfect f1-score of 1.00. This model is also evaluated using a confusion matrix, which can be seen in figure 11, in the table it can be seen that the horizontal part is blue, this shows that the model has succeeded in classifying multi-classes well even though there are some errors. Overall, the model we have created has achieved good performance for brain tumor classification, this can be proven by several evaluation methods that we have shown below.

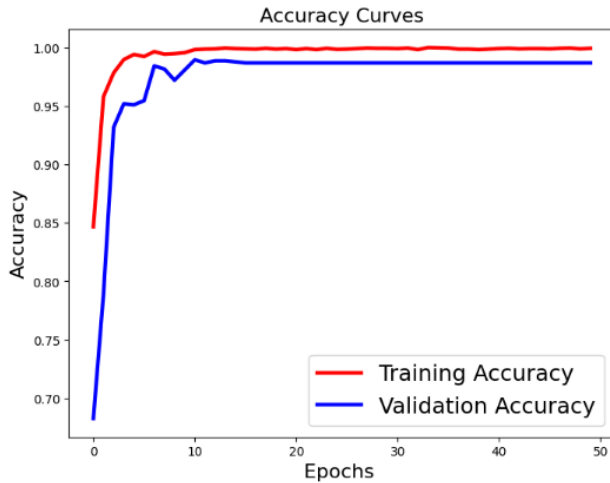


Figure 9. Training and validation accuracy curves of the DenseNet121 model.

	Precision	Recall	f1-score
glioma	0.99	0.97	0.98
meningioma	0.98	0.97	0.98
no tumor	0.99	1.00	1.00
pituitary	0.98	0.99	0.99

Figure 10. Precision, Recall, and F1-score of the DenseNet121 model.

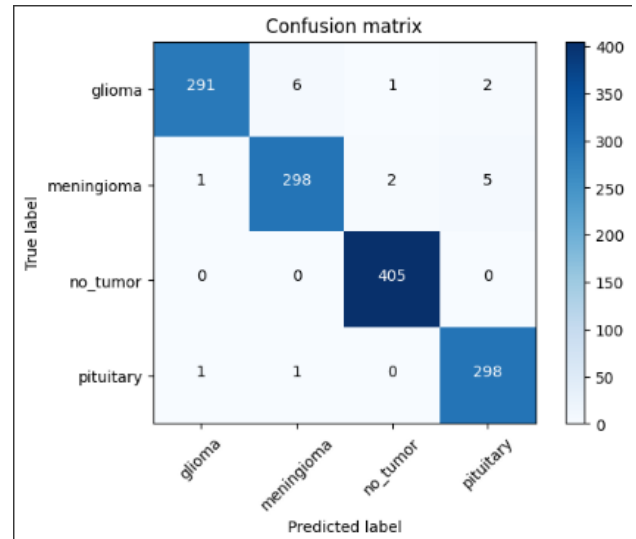


Figure 11. Confusion Matrix of the DenseNet121 model.

4) EfficientNetB3

This EfficientNetB3 model is successful in brain tumor classification of four classes of brain tumors based on accuracy, evaluation metrics, and confusion metrics. The image in figure 12 shows a graph of training and validation accuracy with the number of epochs as the x-axis, shows that in the initial epoch the model experienced quite significant fluctuations in epochs 8-14 and began to stabilize in epoch 15 and then stabilizes near 100%, both on training and validation data, indicating good convergence without significant overfitting. The second figure shows the evaluation metrics in the form of precision, recall, and f1-score for each class with f1 scores above 0.97 in all classes, indicating very high and balanced classification performance. This model is also evaluated using a confusion matrix, which can be seen in figure 14, in the table it can be seen that the horizontal part is blue, this shows that the model has succeeded in classifying multi-classes well even though there are some errors. Overall, the model we have created has achieved good performance for brain tumor classification, this can be proven by several evaluation methods that we have shown below.

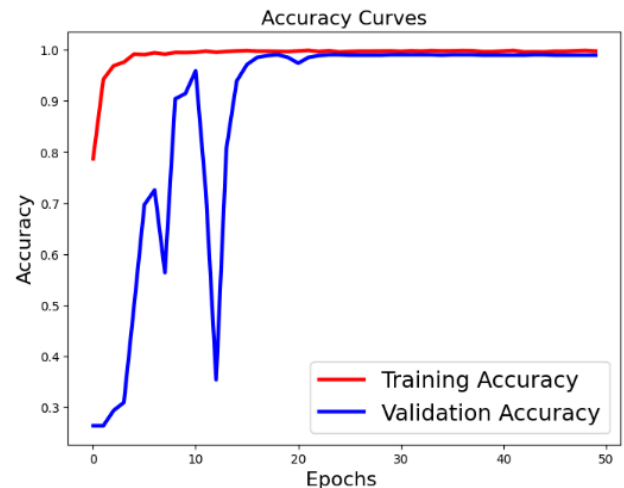


Figure 12. Training and validation accuracy curves of the EfficientNetB3 model.

	Precision	Recall	f1-score
glioma	1.00	0.97	0.98
meningioma	0.97	0.99	0.98
no tumor	0.99	1.00	1.00
pituitary	0.99	0.99	0.99

Figure 13. Precision, Recall, and F1-score of the EfficientNetB3 model.

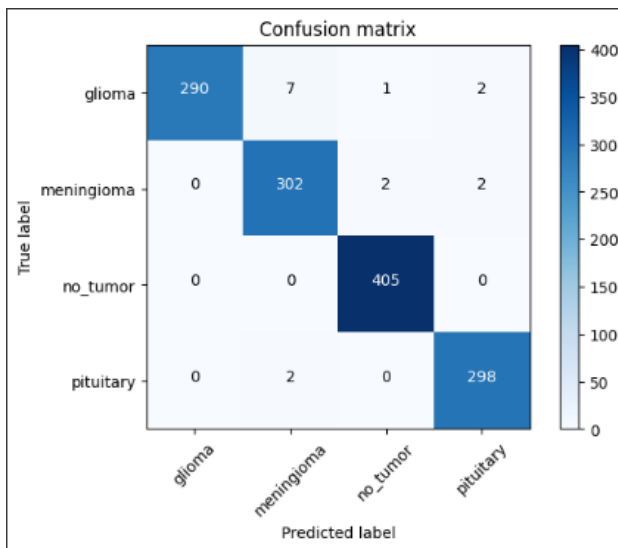


Figure 14. Confusion Matrix of the EfficientNetB3 model.

5) VGG16

In The VGG16 model, the accuracy curve graph, it can be seen that the training accuracy increases significantly to nearly 100%, while the validation accuracy reaches around 97% and is relatively stable after the 10th epoch, indicating that this model has good generalization ability with minimal risk of overfitting. The evaluation metrics table shows high and balanced precision, recall, and f1-score values for all classes, with the highest f1 score in the no_tumor class of 1.00, and other scores ranging from 0.98 to 0.99. This model is also evaluated using a confusion matrix, which can be seen in figure 17, in the table it can be seen that the horizontal part is blue, this shows that the model has succeeded in classifying multi-classes well even though there are some errors. Overall, the model we have created has achieved good performance for brain tumor classification, this can be proven by several evaluation methods that we have shown below.

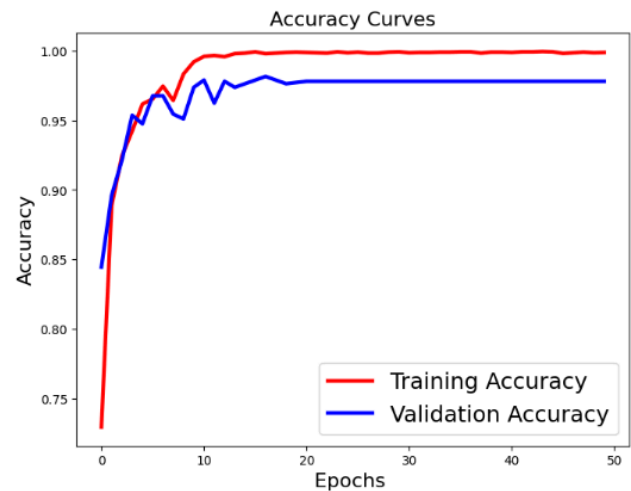


Figure 15. Training and validation accuracy curves of the VGG16 model.

	Precision	Recall	f1-score
glioma	0.99	0.97	0.98
meningioma	0.98	0.99	0.99
no tumor	0.99	1.00	1.00
pituitary	0.98	0.99	0.99

Figure 16. Precision, Recall, and F1-score of the VGG16 model.

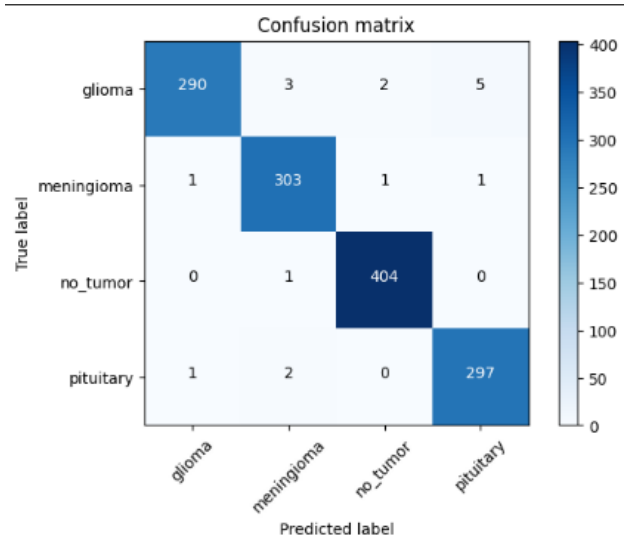


Figure 17. Confusion Matrix of the VGG16 model.

6) Average Drop

Figure above shows the comparison of Average Drop between different deep learning models using Grad-CAM explainability method. Model

InceptionV3 got the highest average drop which is 70.5%, indicating that the model highly relies on the highlighted regions when making predictions. On the other hand, ResNet50 has the lowest score at 40.1%, suggesting that its prediction is less affected by the masked important regions. DenseNet121 also performed quite high at 62.1%, while VGG16 and EfficientNetB3 had more moderate results, around 46-48%. This result suggests that different architectures respond differently to explainability perturbation.

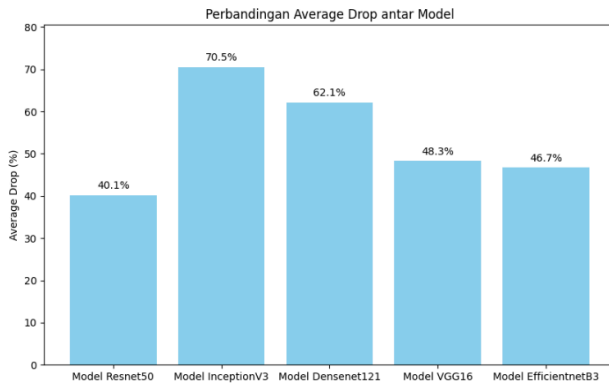


Figure 18. Graphic Average Drop each Models

b. Discussion

Overall, the 5 models that we use for multi-class brain tumor classification have performed well. Therefore, to determine which model is suitable for good brain tumor classification, we use the average drop metric which will measure how much the prediction accuracy decreases when important parts of the image identified by Grad-CAM are removed or removed. This metric provides an overview of the extent to which the model relies on areas that are considered important in decision making, so that models with lower average drop values can be considered to have better interpretability and reliability in brain tumor classification. Based on the graph, it can be seen that the ResNet50 Model is the most stable, with the lowest Average Drop (40.1%). While the inceptionv3 Model experienced the highest performance decline, indicating that this model is the least stable against testing conditions that cause average Drop.

V. Conclusion and Suggestions

This study has successfully implemented and evaluated five pre-trained CNN models—ResNet50, VGG16, EfficientNetB3, DenseNet121, and InceptionV3—to classify brain tumor types from MRI images and to analyze their interpretability using Grad-CAM. All models demonstrated high classification performance, with DenseNet121 and EfficientNetB3 achieving the highest accuracy and generalization ability, while ResNet50 showed the best interpretability based on the lowest Average Drop value (40.1%).

Key findings include:

- All models achieved over 95% accuracy, with DenseNet121 and EfficientNetB3 nearly reaching 99%.
- Grad-CAM visualizations confirmed that models are focusing on clinically relevant areas, aiding interpretability.
- Inference speed was fastest in ResNet50 and VGG16, but slightly slower in more complex models like EfficientNetB3 and InceptionV3.
- Average Drop analysis indicated that ResNet50 provides the most stable and interpretable predictions.

Suggestions

- Ensemble Model Integration:** Future work should explore integrating these models using ensemble methods (e.g., stacking, majority voting) to leverage their individual strengths.
- Clinical Validation:** Collaborations with medical professionals and radiologists are necessary to further validate model predictions and Grad-CAM heatmaps with real-world clinical data.
- Lightweight Deployment:** Optimization for deployment on edge devices (e.g., mobile or embedded systems) can be explored using model compression techniques.
- Explainability Enhancement:** Combining Grad-CAM with other explainability tools like LIME or SHAP could further enhance trust in AI decisions.
- Dataset Expansion:** Applying these models to larger and more diverse datasets could test generalization capabilities across different patient populations and MRI scanners.

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